

MineArt Diffusion

Project Write-Up — Generative AI, Computer Vision & Data Visualization

1. Project Title

MineArt Diffusion — A machine-learning-based system for generating Minecraft-style artwork from text prompts and input images, followed by visual analysis and Minecraft-compatible export.

2. Problem Statement

Creating customized Minecraft artwork traditionally requires manual pixel-art or image-editing skills. Existing generative image systems are not specifically optimized for Minecraft visual characteristics and generally do not provide detailed analysis of the generated artwork. There is a need for a specialized system that can generate Minecraft-style images while also demonstrating the underlying machine-learning process, quantitative evaluation, and visual composition analysis.

3. Proposed System

MineArt Diffusion will use a custom-trained diffusion model and a fine-tuned pretrained diffusion model specialized using a self-created Minecraft screenshot dataset. Users will be able to enter a text prompt or provide an image as input. The generated image will then be analyzed using computer vision and semantic segmentation to estimate the contribution of elements such as sky, land, clouds, mountains, buildings, and vegetation. Interactive visualizations will present dataset statistics, diffusion behavior, training results, and model evaluation. The final artwork will be processed for Minecraft-compatible poster/painting export.

4. Objectives

- Build and preprocess a large Minecraft image dataset using automated tools.
- Implement and train a diffusion model using PyTorch.
- Fine-tune a pretrained diffusion model for Minecraft-style generation.
- Apply mathematical concepts such as noise schedules, probability distributions, optimization, and color quantization.
- Develop a semantic-segmentation pipeline for generated-image analysis.
- Evaluate models using metrics such as FID, KID, CLIP similarity, and segmentation IoU where appropriate.
- Build interactive data visualizations for the dataset, training process, and model performance.
- Provide a full-stack application for generation, analysis, and Minecraft export.

5. Scope

The project covers dataset engineering, custom diffusion-model training, pretrained-model fine-tuning, image-to-image generation, computer-vision analysis, model evaluation, applied mathematics, interactive visualization, backend APIs, and a web-based user interface. Advanced features such as image-plus-text conditioning, diffusion-step exploration, embedding visualization, and composition-aware regeneration may be added after the core system is functional.

6. Technology Stack

Machine Learning: Python, PyTorch, Hugging Face Diffusers, PEFT/LoRA, CUDA

Data & Computer Vision: NumPy, Pandas, OpenCV, Pillow, scikit-learn

Experiment Tracking: MLflow

Frontend: React, TypeScript, Vite, Tailwind CSS, Apache ECharts

Backend: Node.js, TypeScript, Fastify

Infrastructure: PostgreSQL, Redis, Celery, MinIO/S3-compatible storage, Docker

7. Expected Outcomes

The project is expected to produce a working Minecraft-focused generative AI system, a self-created and reproducible training dataset, a custom-trained diffusion model, a Minecraft-specialized fine-tuned model, and a computer-vision analysis pipeline. The final application will provide quantitative dashboards showing dataset

characteristics, training behavior, diffusion processes, model metrics, and image composition. Users will also be able to export generated artwork in a Minecraft-oriented poster/painting format.

8. Student Name

Student: Akshat Pratap Singh Gaur

9. Timeline

A phased development plan will be followed: **Phase 1:** Dataset collection and automated preprocessing → **Phase 2:** Custom DDPM implementation and training → **Phase 3:** Training experiments and evaluation → **Phase 4:** Pretrained-model fine-tuning → **Phase 5:** Segmentation and image analysis → **Phase 6:** Data-visualization dashboard → **Phase 7:** Full-stack integration and Minecraft export → **Phase 8:** Testing, optimization, documentation, and final demonstration.

Primary Academic Areas	Project Contribution
Machine Learning	Diffusion model training, fine-tuning, segmentation, evaluation
Data Visualization	Dataset, training, diffusion, segmentation and metric dashboards
Applied Mathematics	Diffusion equations, probability, optimization, IoU and color quantization
Software Engineering	Modular backend, workers, database, storage, testing and versioning
Full-Stack Development	React frontend, FastAPI backend and ML-service integration